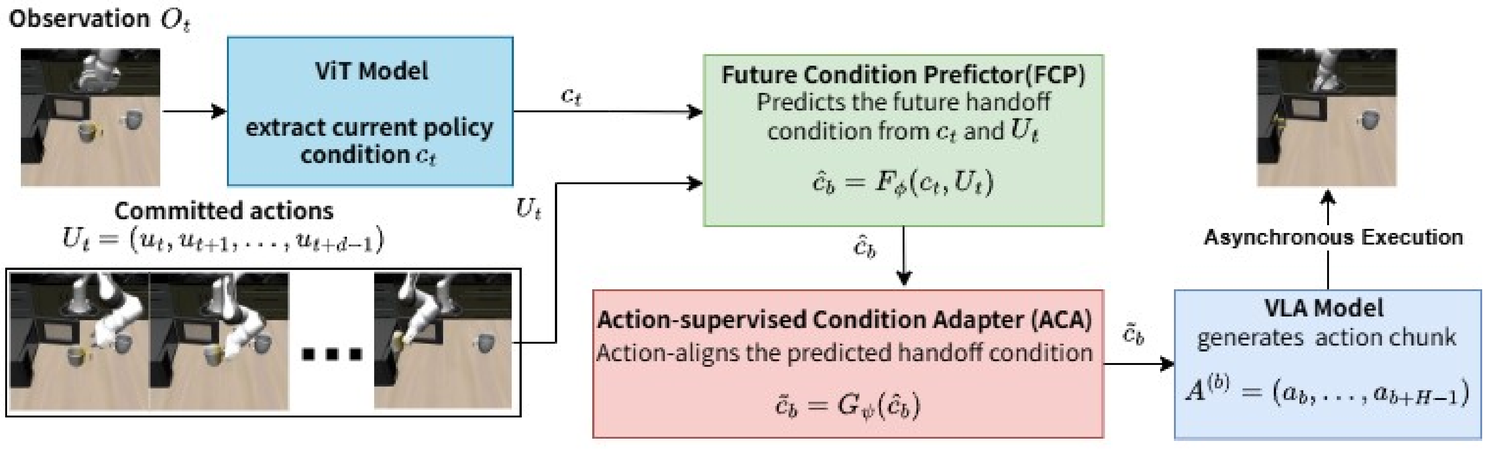
A-PACE: Asynchronous Predictive Alignment and Continuity for Frozen VLA Execution

Key Insight

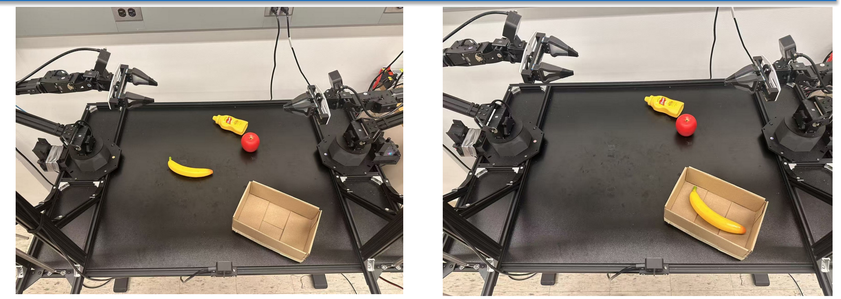
Inference latency creates observation–execution mismatch
A-PACE predicts the **future handoff condition** and performs **action-supervised condition alignment**

A-PACE Framework

FCP predicts the future handoff condition from the query-time condition and committed actions. ACA aligns the prediction with the frozen VLA.



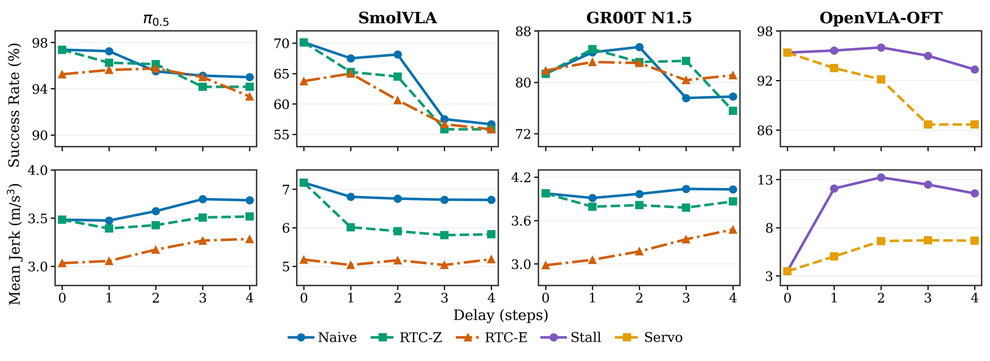
Real World Experiments



(a) Initial Configuration (b) Successful Placement

Metric	Naive	RTC	A-PACE w/o ACA	A-PACE
Success (%) ↑	53.3	56.7	53.3	**63.3**
Jerk (S) ↓	9.20	**4.95**	7.07	7.13
Jerk (F) ↓	8.11	**4.81**	7.69	6.98

Simulation Experiments on LIBERO



Model	Method	Success (%) ↑	Mean EE Jerk ↓
SmolVLA	A-PACE	**73.17**	7.124
	A-PACE w/o ACA	71.17	7.301
	FutureRTC	40.17	7.841
	RTC-Exp	53.33	**5.661**
	Naive Async	50.00	6.840
$\pi_{0.5}$	A-PACE	**97.00**	3.437
	A-PACE w/o ACA	96.17	3.475
	FutureRTC	95.50	3.522
	RTC-Exp	89.17	**3.401**
	Naive Async	85.83	3.895

A-PACE achieves the highest success on both VLA backbones

Simulation Experiments on kinetix

